



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

UNIT TEST

12-06-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TNAS)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C,D	A,B,C,D	A,D	A,B,C,D	A,B,C	A,C	A,B,C,D	C,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	A,B,D	A,C	A,B,C	A,B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	9.00	2.00	0.38 to 0.39	0.25 to 0.26	3.00	4.00	6.66 to 6.67	3.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,D	B,C,D	A,C	C,D	A,B,C	A	A,C	B,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	A,C	B	A,B,C,D	A,C,D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	5.00	8.00	1.00	9.45	4.27	9.00	3.00	4.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	C	B	B,C,D	B,C	A	A,B,C,D	A,C	A,B,C
SECTION-I (ii)	Q.	9	10	11	12				
	A.	B	A,B,C	B,C,D	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.50	6.00	8.00	4.58 to 4.59	0.03	1.00	3.00	0.50

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B,C,D)

When only S_2 is open

$$\frac{2}{3} = \left(\frac{4}{120}\right) \ell \Rightarrow \ell = 20\text{cm}$$

When only S_1 is open

$$2 = \left(\frac{4}{120}\right) \ell \Rightarrow \ell = 60\text{cm}$$

When all three are closed

$$2 = \frac{4}{120} \ell \Rightarrow \ell = 60\text{cm}$$

When only S_1 is closed

$$2 = \frac{4}{120} \times \ell \Rightarrow \ell = 60\text{cm}$$

2. Ans (A,B,C,D)

The point P will trace a curve given by

$$\frac{x^2}{\left(\frac{\ell}{2} - r\right)^2} + \frac{y^2}{r^2} = 1 \dots (1)$$

$$\text{For } r = \frac{\ell}{4}, \frac{x^2}{\left(\frac{\ell}{4}\right)^2} + \frac{y^2}{\left(\frac{\ell}{4}\right)^2} = 1 \text{ or } x^2 + y^2 = \left(\frac{\ell}{4}\right)^2$$

$\Rightarrow$ Quadrant of a circle

Taking moments of forces about the point A'

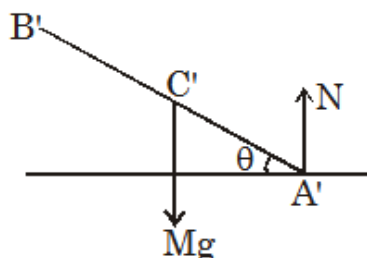
$$Mg \cdot \frac{\ell}{2} \cos \theta = -I\theta, I = \frac{M\ell^2}{3}, \text{ moment of inertia about A'}$$

$$\ddot{\theta} = -\frac{3g}{2\ell} \cdot \cos \theta \text{ [-ve sign : } \theta \text{ decreases as } t \text{ increases]}$$

$$= \frac{d\omega}{dt}, \omega = \frac{d\theta}{dt}$$

$$\therefore \frac{d\omega}{d\theta} \cdot \omega = -\frac{3g}{2\ell} \cos \theta$$

$$\text{Integrating, } \frac{\omega^2}{2} = -\frac{3g}{2\ell} \sin \theta + C$$



$$\text{At } \theta = \frac{\pi}{2}, \omega = 0 \text{ hence } C = \frac{3g}{2\ell}$$

$$\omega^2 = \frac{3g}{\ell} (1 - \sin \theta)$$

3. **Ans (A,D)**

The moment of inertia of a uniform equilateral lamina of mass M and an edge length of a about an axis through its centroid O perpendicular to its plane is given by $I = \frac{Ma^2}{12}$. Hence the moment of inertia of the system about an axis through O perpendicular to its plane is

$$I_s = I + 3 \left(\frac{M}{3} \cdot \left(\frac{a}{2\sqrt{3}} \right)^2 \right) = \frac{Ma^2}{12} + \frac{Ma^2}{12} = \frac{Ma^2}{6}$$

Similarly by theorem of parallel axes, applied to the two parallel axes through A and O , both perpendicular to plane is

$$I_A = I_0 + (2M)d^2 \text{ where } d \text{ is the distance between the axes.}$$

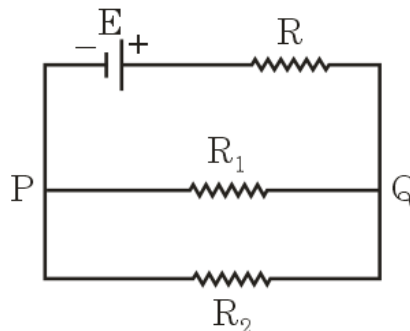
$$\text{Now } I_0 = \frac{Ma^2}{6}, \text{ and } d = \frac{2}{3} \left[\frac{\sqrt{3}a}{2} \right] = \frac{a}{\sqrt{3}}$$

Hence

$$I_A = \frac{Ma^2}{6} + (2M) \frac{a^2}{3} = Ma^2 \left[\frac{1}{6} + \frac{2}{3} \right] = \frac{5Ma^2}{6}$$

This is also the M of I about axis through C .

5. **Ans (A,B,C)**



(A) If $P_1 = P_2$, $\frac{V^2}{R_1} = \frac{V^2}{R_2}$ where V is the difference of potential from P to Q . Hence $R_1 = R_2$

(B) $P = i^2 R = i^2 (R_1 + R_2) = i^2 R_1 + i^2 R_2 > I_1^2 R_1 + I_2^2 R_2$ since $i > i_1$ and $i > i_2$

Hence $P > P_1 + P_2$

(C) Let $P = P_1 + P_2$ then $i^2 R = i_1^2 R_1 + i_2^2 R_2$

$$\text{But } i = i_1 + i_2 \Rightarrow (i_1 + i_2)^2 R = i_1^2 R_1 + i_2^2 R_2$$

$$\text{Now } (i_1 + i_2)^2 = (i_1 - i_2)^2 + 4i_1 i_2$$

Since i_1 and i_2 are both positive, maximum value of $(i_1 + i_2)$ is when $i_1 = i_2 = \frac{i}{2}$

$$\text{Then from (1), } 4R = R_1 + R_2$$

Hence maximum value R can have consistent with

$$P = P_1 + P_2 \text{ is } \frac{R_1 + R_2}{4}$$

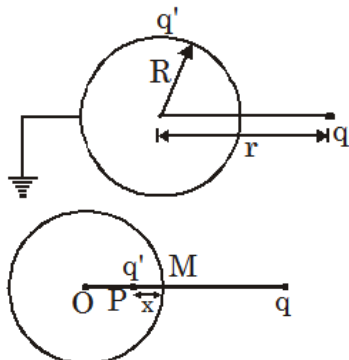
$$\text{Hence } R \leq \frac{R_1 + R_2}{4}$$

6. Ans (A,C)

Potential of sphere is zero

$$\frac{kq'}{R} + \frac{kq}{r} = 0$$

$$\Rightarrow q' = -\left(\frac{R}{r}\right)q$$



We can assume whole charge q' is present at point P which is at distance $(R - x)$ from centre of sphere.

We know potential of point M is same as centre so

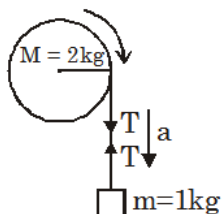
$$\frac{kq'}{x} + \frac{kq}{(r - R)} = 0 ; x = \frac{R(r - R)}{r}$$

So force is

$$F = \frac{1}{4\pi\epsilon_0} \frac{qq'}{(r - R + x)^2} = \frac{1}{4\pi\epsilon_0} \frac{q^2 Rr}{(r^2 - R^2)^2}$$

7. Ans (A,B,C,D)

By FBD of particle



$$mg - T = ma$$

$$10 - T = a \dots\dots(i)$$

By FBD of disc

$$TR = I\alpha = L \frac{a}{R} \Rightarrow T = \frac{MR^2}{2} \frac{a}{R^2}$$

$$T = Ma/2 = a \dots\dots(ii)$$

By eq. (i) and (ii)

$$(A) a = 5 \text{ m/s}^2 \text{ and } T = 5 \text{ N and } \alpha = a/R = 5 \text{ rad/s}^2$$

(B) For angular displacement of disc :

$$\theta = \omega t + 1/2 \alpha t^2$$

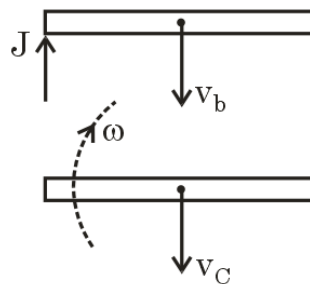
(C) Work done by torque

$$= \int \tau d\theta = \tau \int d\theta = 5 \times 40 = 200 \text{ J}$$

$$(A) \Delta K.E. = \Delta w = 200 \text{ J}$$

$$K_2 - K_1 = 200 \text{ J}$$

8. Ans (C,D)



$$mv_0 - mv_c = J \dots\dots(1)$$

$$J \frac{l}{2} = \frac{m l^2}{12} \omega \text{ or } J = \frac{m l \omega}{6} \dots\dots(2)$$

$$v_0 - v_c = \frac{l \omega}{6}$$

$$3v_0 - 3v_c = \frac{l \omega}{2} \dots\dots(3)$$

$$e = \frac{v_c - \frac{l \omega}{2}}{v_0} \Rightarrow v_c = e v_0 + \frac{l \omega}{2} \dots\dots(4)$$

$$(3) \& (4) \Rightarrow v_c = e v_0 + 3v_0 - 3v_c$$

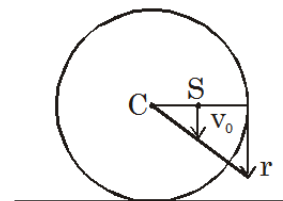
$$\Rightarrow v_c = \frac{v_0}{4} (3 + e)$$

n_c is always +ve for all values of 'e'

PART-1 : PHYSICS

SECTION-I (ii)

9. Ans (A,B,D)



In the sequence of horizontal external force centre of mass of system S falls vertically. Since mass of ring and particle are equal their motion about CM of system is symmetrical. At given instant centre of ring will be instantaneous centre.

From work energy theorem

$$mgr = \frac{1}{2} m v_1^2 + \frac{1}{2} m r^2 \omega^2$$

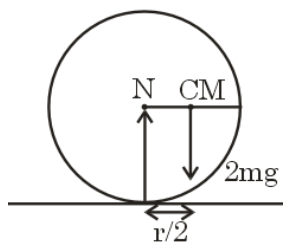
$$\text{also } v_1 = r\omega$$

$$\text{and } v_1 = \sqrt{gr}$$

$$|\vec{a}_c| = |\vec{a}_i| = \frac{v_{\text{rel}}^2}{(r/2)} = \frac{(v_i/2)^2}{r/2} = \frac{g}{2}$$

$$\left[v_{\text{re}} = v_i - v_s = \frac{v_1}{2} \right]$$

At the given instant acceleration of particle is equal to the centripetal acceleration of its rotation about centre of mass.



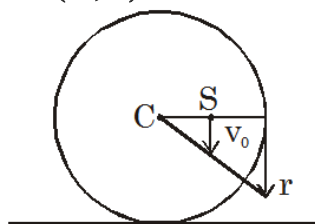
$$N \frac{r}{2} = \frac{3}{2} m r^2 \alpha \quad \dots(i)$$

$$a_s = \frac{r}{2} \alpha \quad \dots(ii)$$

$$a_s = \frac{2mg - N}{2m} \quad \dots(iii)$$

$$\text{Thus } N = \frac{3}{2} mg$$

10. **Ans (A,C)**



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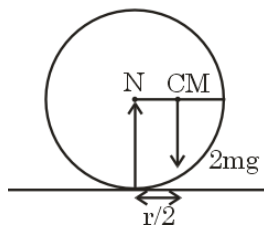
$$\text{also } v_1 = r\omega$$

$$\text{and } v_1 = \sqrt{gr}$$

$$|\vec{a}_c| = |\vec{a}_i| = \frac{v_{rel}^2}{(r/2)} = \frac{(v_1/2)^2}{r/2} = \frac{g}{2}$$

$$[v_{re} = v_i - v_s = \frac{v_1}{2}]$$

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$$N \frac{r}{2} = \frac{3}{2} m r^2 \alpha \quad \dots(i)$$

$$a_s = \frac{r}{2} \alpha \quad \dots(ii)$$

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$$\text{Thus } N = \frac{3}{2} mg$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (9.00)**

$$W = \Delta KE$$

$$W_g + W_E = \Delta KE$$

$$(1) W_g = \frac{-1}{2} m 125$$

$$(2) W_g + W_E = \frac{-1}{2} m (13)^2$$

$$(3) W_g - W_E = \frac{-1}{2} m v^2$$

$$2W_g = \frac{-1}{2} m (13)^2 - \frac{1}{2} m v^2$$

$$\frac{1}{2} \times m (125) = m (13)^2 + m v^2$$

$$v = 9$$

2. **Ans (2.00)**

$$W = q\Delta V = q\alpha \frac{a}{2} (2a) = \frac{1}{2} m v^2$$

$$v^2 = \frac{2q\alpha a^2}{m} = \frac{2 \frac{2m}{a^2} a^3}{m}$$

$$v = 2$$

3. **Ans (0.38 to 0.39)**

$$U = \frac{1}{2} \sum q_i v_i$$

$$U = \frac{1}{2} \left[Q \left(\frac{kQ}{a} - \frac{kQ}{2a} + \frac{k2Q}{3a} + \frac{kQ}{4a} \right) \right]$$

$$+ Q \left(\frac{kQ}{2a} - \frac{kQ}{2a} + \frac{k2Q}{3a} + \frac{kQ}{4a} \right)$$

$$+ 2Q \left(\frac{kQ}{3a} - \frac{kQ}{3a} + \frac{k2Q}{3a} + \frac{kQ}{4a} \right)$$

$$+ Q \left(\frac{kQ}{4a} - \frac{kQ}{4a} + \frac{k2Q}{4a} + \frac{kQ}{4a} \right)$$

$$U = \frac{1.54kQ^2}{a}$$

4. **Ans (0.25 to 0.26)**

$$G = 50$$

$$R = 1150$$

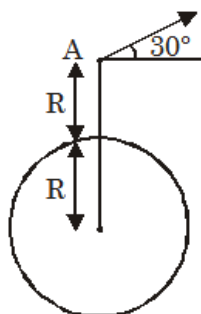
$$V = i_g (R + G)$$

$$i_z = i_g 1200$$

$$i_g = \frac{1}{100}$$

$$S = \frac{i_g G}{i - i_g} \approx 0.25$$

5. Ans (3.00)



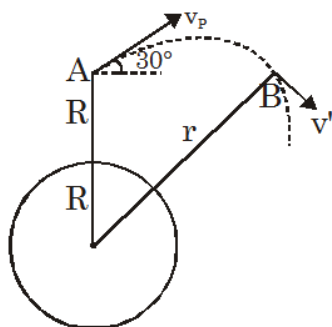
Let v_0 be the velocity required to send a particle to infinity from point A.

$$\frac{1}{2}mv_0^2 - \frac{GMm}{2R} = 0$$

$$\Rightarrow v_0 = \sqrt{\frac{GM}{R}}$$

$\therefore$ Velocity of projection at A

$$v_p = \frac{1}{\sqrt{2}}v_0 = \sqrt{\frac{GM}{2R}}$$



Let V be the velocity at the highest point and r be its distance from the center of the planet. Using conservation of angular momentum.

$$mv_p 2R \cos 30^\circ = mvr$$

$$\Rightarrow v = \frac{\sqrt{3}R}{r} \sqrt{\frac{GM}{R}} = \sqrt{\frac{3GMR}{2r^2}} \quad \dots\dots\dots(1)$$

From conservation on the mechanical energy at A and B

$$-\frac{GMm}{2R} + \frac{1}{2}m \cdot \frac{GM}{2R} = -\frac{GMm}{r} + \frac{1}{2}mv^2$$

replacing the value of V from (1) in the above equation.

$$-\frac{GMm}{2R} + \frac{1}{2}m \cdot \frac{GM}{2R} = -\frac{GMm}{r} + \frac{1}{2}mv'^2$$

replacing the value of v' from (1) in the above equation.

$$-\frac{GM}{4R} = -\frac{GM}{r} + \frac{1}{2} \times \frac{3}{2} \frac{GMR}{r^2} \Rightarrow r^2 - 4Rr + 3R^2 = 0$$

$$\Rightarrow (r - 3R)(r - R) = 0 \Rightarrow r = R, r = 3R$$

6. Ans (4.00)

For the element

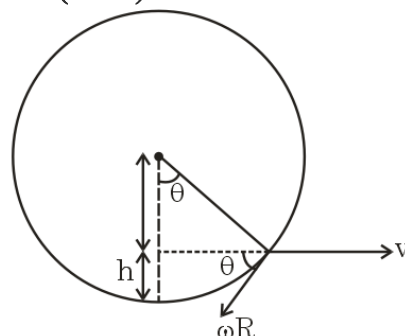
$$i = \frac{dV}{\frac{1}{\sigma} \cdot \frac{dx}{4\pi^2}} = \frac{Edx}{\frac{1}{KE} \cdot \frac{dx}{4\pi^2}}$$

$$i = KE^2 4\pi x^2$$

$$\Rightarrow \sqrt{\frac{i}{4\pi K}} = E \cdot x = \frac{dV}{dx} \cdot x \Rightarrow \sqrt{\frac{1}{4\pi K}} \int_a^b \frac{dx}{x} = \int_0^V dV$$

$$\Rightarrow i = \frac{4\pi K V^2}{\left(\ln \frac{b}{a}\right)^2} = \frac{4\pi \times \frac{1}{\pi} \times 10^{-2} \times (20)^2}{(1)^2} = 16$$

8. Ans (3.00)



$$v = \omega R \cos \theta$$

$$\cos \theta = \frac{2}{5}$$

$$h = R - R \cos \theta$$

$$h = 3 \text{ cm}$$

PART-2 : CHEMISTRY

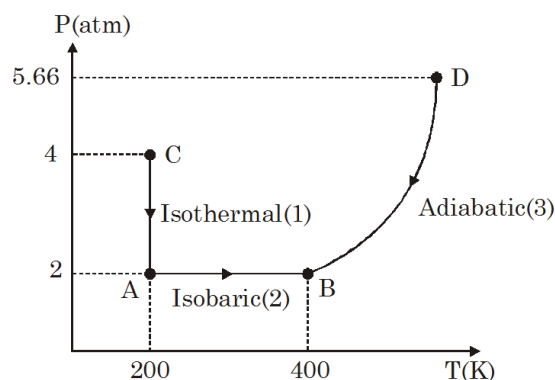
SECTION-I (i)

4. Ans (C,D)

For adiabatic process,

$$P_1 V_1^\gamma = P_B V_B^\gamma \Rightarrow P_1 = 2 \times \left(\frac{V_B}{V_1}\right)^{1.5}$$

$$= 2 \times 2^{1.5} = 5.66 \text{ atm}$$



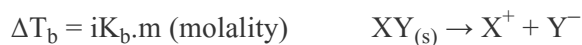
PART-2 : CHEMISTRY

SECTION-II

3.

Ans (1.00)

Elevation in boiling point Assuming ionic compound to be xy



$$0.52 = i \cdot 0.52 \times \frac{\text{moles}}{1 \text{ kg water (solvent)}}$$

i = Van'toff factor

$$\text{moles of compound} = \frac{1}{2} \text{ mole} = 2$$

$$\frac{1}{2} \text{ mole compound} = 3 \times 10^{23} \text{ molecules as}$$

100% dissociation

1 unit cell like fcc (NaCl) contains 4 moles of ionic compound

$$\text{So, number of unit cells} = \frac{3 \times 10^{23}}{4}$$

$$r^- = \text{radius of anion} = 2 \text{ \AA}$$

$$r^+ = \text{radius of cation} = r^+ \text{ \AA}$$

$$\text{edge length} = 2(r^- + r^+) = 2(2 + r^+) \text{ \AA}$$

$$\text{Volume of solid cube} = (2.532)^3 \text{ cm}^3$$

$$\text{Volume of all the unit cell} =$$

$$\frac{3}{4} \times 10^{23} \times [2(2 + r^+)]^3$$

No. of unit cell

$$21.6 = 10^{23} \times (2(2 + r^+))^3$$

$$6 \times 10^{-8} \text{ cm} = 2(2 + r^+) \text{ \AA}$$

$$6 \times 10^{-8} \text{ cm} = 2(2 + r^+) 10^{-8} \text{ cm}$$

4. Ans (9.45)

5. Ans (4.27)

$$k_1 \rightarrow k_1' \quad k_1' = \text{including stoichiometry}$$

$$k_2 \rightarrow k_2'$$

$$k_3 \rightarrow k_3'$$

$$K_1' : K_2' : K_3' :: 2 : 3 : 4$$

$$K_{\text{net}} = A e^{-E_a/RT}$$

$$K = A e^{-E_a/RT}$$

$$A e^{-E_a/RT} = A_1 e^{-E_{a1}/RT} + A_2 e^{-E_{a2}/RT} + A_3 e^{-E_{a3}/RT}$$

$$\ln K = -\frac{E_a}{RT} + \ln(A)$$

$$\ln(A e^{-E_a/RT}) = \ln(A_1 e^{-E_{a1}/RT} + A_2 e^{-E_{a2}/RT} + A_3 e^{-E_{a3}/RT})$$

$$\frac{\Delta K}{K} = \frac{-\Delta E_a}{RT}$$

$$-\frac{\Delta E_a}{RT} = \frac{\Delta(K_1 + K_2 + K_3)}{K_1 + K_2 + K_3}$$

$$-\frac{\Delta E_a}{RT} = \frac{-\frac{\Delta E_1}{RT} K_1 + -\frac{\Delta E_2}{RT} K_2 + -\frac{\Delta E_3}{RT} K_3}{K_1 + K_2 + K_3}$$

$$\Delta E_a = \frac{\Delta E_{a1} K_1 + \Delta E_{a2} K_2 + \Delta E_{a3} K_3}{K_1 + K_2 + K_3}$$

$$E_{a_{\text{net}}} = \frac{K_1 E_{a1} + K_2 E_{a2} + K_3 E_{a3}}{K_1 + K_2 + K_3}$$

$$K_1' : K_2' : K_3' :: 2 : 3 : 4$$

$$K_1' = 3K_1 \quad 3K_1 = 2K_2 \quad K_1 : K_2 : K_3$$

$$K_2' = 2K_2 \quad 2K_2 = 3K_3 \quad \frac{2}{3} : \frac{3}{2} : 4$$

$$K_3' = K_3 \quad K_3 = 4K_1 \quad K_1 : K_2 : K_3$$

$$4 : 9 : 24$$

Using these values =

$$\frac{4 \times 8 + 9 \times 6 + 24 \times 3}{4 + 9 + 24} = 4.27 \text{ Kcal}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (C)**

$$\begin{aligned} \lim_{x \rightarrow \infty} x^2 \tan \left(\ln \left(\cos \frac{\pi}{x} \right)^{1/2} \right) &= \lim_{x \rightarrow \infty} x^2 \left(\ln \left(\cos \frac{\pi}{x} \right)^{1/2} \right) \\ &= \lim_{x \rightarrow \infty} \frac{x^2}{2} \ln \left(\cos \frac{\pi}{x} \right) \\ &= \lim_{x \rightarrow \infty} \frac{x^2}{2} \left(\ln \left(1 + \left(\cos \frac{\pi}{x} - 1 \right) \right) \right) = \lim_{x \rightarrow \infty} \frac{x^2}{2} \left(\cos \frac{\pi}{x} - 1 \right) \\ &= \lim_{x \rightarrow \infty} \frac{x^2}{2} \left(-2 \sin^2 \frac{\pi}{2x} \right) = -\frac{\pi^2}{4} \end{aligned}$$

2. **Ans (B)**

$$\int \frac{x^3 - 8}{x^3 \sqrt{x^4 - x^3 + 2}} dx = \int \frac{\frac{1}{x^2} - \frac{8}{x^5}}{\sqrt{1 - \frac{1}{x} + \frac{2}{x^4}}} dx = \frac{2}{x^2} \sqrt{x^4 - x^3 + 2} + C$$

$\therefore k = 2$

3. **Ans (B,C,D)**

Differentiate on both side

$$\begin{aligned} \frac{3z^3 - 8z + 5}{\sqrt{z^2 - 4z - 7}} \\ &= (2z + a) \sqrt{z^2 - 4z - 7} + \frac{(z^2 + az + 36)}{\sqrt{z^2 - 4z - 7}} (z - 2) + \frac{b}{\sqrt{z^2 - 4z - 7}} \\ &\Rightarrow 3z^3 - 8z + 5 \\ &= (2z + a)(z^2 - 4z - 7) + (z^2 + az + 36)(z - 2) + b \\ &\text{equating the coefficient of } z^2 \text{ and constant} \\ a - 8 + a - 2 &= 0 \Rightarrow a = 5 \\ 5 &= -7a - 72 + b \Rightarrow b = 112 \end{aligned}$$

4. **Ans (B,C)**

$$\left[\frac{x+1}{x} > 0 \right] \Rightarrow \left[x \in (-\infty, -1) \cup (0, \infty) \right] \dots (1)$$

$$\left[\left| \ln \left(\frac{x+1}{x} \right) \right| \leq 1 \right] \Rightarrow \left[\frac{1}{e} \leq \frac{x+1}{x} \leq e \right] \dots (2)$$

$$\text{From (2), } \frac{x+1-ex}{x} \leq 0 \text{ and } \frac{ex+e-x}{ex} \geq 0$$

$$\Rightarrow \frac{x(e-1)-1}{x} \geq 0 \text{ and } \frac{x(e-1)+e}{ex} \geq 0$$

$$x \in (-\infty, 0) \cup \left[\frac{1}{e-1}, \infty \right)$$

$$\text{and } x \in \left(-\infty, \frac{e}{1-e} \right] \cup (0, \infty)$$

from (1) and (2)

$$x \in \left(-\infty, \frac{e}{1-e} \right] \cup \left[\frac{1}{e-1}, \infty \right)$$

and range of $f(x)$ is $[-1, 1]$

5. **Ans (A)**

$$f(2-x) = f(2+x)$$

$$x \rightarrow 2-x$$

$$f(x) = f(4-x) \dots (i)$$

differentiate w.r.t. x

$$f'(x) = -f'(4-x) \dots (ii)$$

Put $x = 1/2$ in (ii)

$$f'\left(\frac{1}{2}\right) = f'\left(\frac{7}{2}\right) = f'(1) = f'(2) = f'(3) = 0$$

also

Put $x = 1, 2$

Now consider $f'(x) = y$.

by Roll's theorem we can say in

$$\left[\frac{1}{2}, 1 \right], [1, 2], \left[2, \frac{7}{2} \right], \left[\frac{7}{2}, 3 \right]$$

$f''(x) = 0$ has minimum 4 roots.

6. **Ans (A,B,C,D)**

$$F(n, x) = \sum_{r=1}^n \tan^{-1} \left(\frac{r+1}{2} x \right) - \tan^{-1} \left(\frac{r}{2} \right) x$$

$$= \tan^{-1} \left(\frac{n+1}{2} x \right) - \tan^{-1} \left(\frac{x}{2} \right)$$

$$\Rightarrow G(n, x) = \frac{2(n+1)}{4 + (n+1)^2 x^2} - \frac{2}{4 + x^2}$$

$$G\left(n, \frac{1}{3}\right) = \frac{18(n+1)}{36 + (n+1)^2} - \frac{18}{37}$$

$$G\left(n, \frac{1}{3}\right) < 0 \Rightarrow n^2 > 35n \Rightarrow n \geq 36$$

7. **Ans (A,C)**

$$(a+b) + (b+c) + (a+c) = 8$$

$$\text{let } a+b = \alpha, b+c = \beta, a+c = \gamma$$

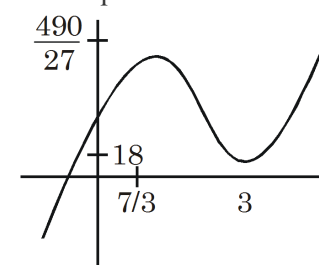
$$\Rightarrow \alpha + \beta + \gamma = 8$$

$$\Rightarrow \alpha\beta + \beta\gamma + \gamma\alpha = a^2 + b^2 + c^2 + 3(ab + bc + ac) = 21$$

$$\Rightarrow x^3 - 8x^2 + 21x - \lambda = 0$$

$$\text{where } \lambda = (a+b)(b+c)(a+c)$$

these equation has 3 real roots

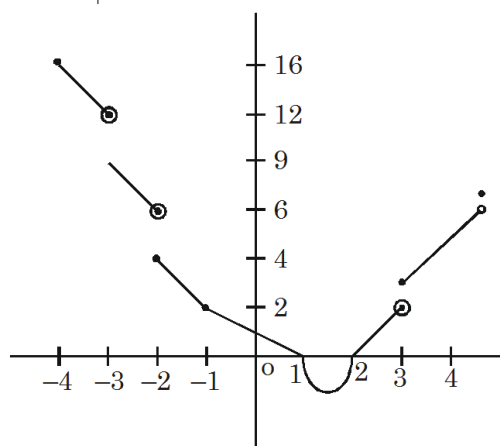


$$\text{so } f(3)f(7/3) \leq 0$$

$$18 \leq \lambda \leq \frac{490}{27}$$

8. Ans (A,B,C)

$$f(x) = \begin{cases} x[x] & -4 \leq x < -1 \\ 1-x & -1 \leq x < 1 \\ x^2 - 3x + 2 & 1 \leq x < 2 \\ (x-2)[x] & 2 \leq x \leq 4 \\ -4x & x \in [-4, -3) \\ -3x & x \in [-3, -2) \\ -2x & x \in [-2, -1) \\ 1-x & x \in [-1, 1) \\ x^2 - 3x + 2 & x \in [1, 2) \\ 2(x-2) & x \in [2, 3) \\ 3(x-2) & x \in [3, 4) \\ 8 & x = 4 \end{cases}$$



- (1) $f(x)$ is DC at $x = -3, 2, 3, 4 \rightarrow 4$ points
- (2) $f(x)$ is ND at $x = -3, -2, 3, 4, -1, 2 \rightarrow 6$ points
- (3) minima at $x = \frac{3}{2}$, maxima at $x = -4, 4$
- (4) no. of solution of $f(x) = 4$ is 2
- (5) Sum of global maximum value and global minimum value of $\frac{63}{4}$

PART-3 : MATHEMATICS

SECTION-I (ii)

9. Ans (B)

$$f(x) = \frac{1}{2} \left[f(xy) + f\left(\frac{x}{y}\right) \right] \quad \dots(i)$$

interchange x and y

$$f(y) = \frac{1}{2} \left[f(xy) + f\left(\frac{y}{x}\right) \right] \quad \dots(ii)$$

Now (i) - (ii)

$$f(x) - f(y) = \frac{1}{2} \left\{ f\left(\frac{x}{y}\right) - f\left(\frac{y}{x}\right) \right\}$$

Put $x = 1$,

$$f(1) = \frac{1}{2} \left\{ f(y) + f\left(\frac{1}{y}\right) \right\}$$

$$\Rightarrow f(1) = 0 \Rightarrow f(y) = -f\left(\frac{1}{y}\right)$$

$$\Rightarrow f\left(\frac{x}{y}\right) = -f\left(\frac{y}{x}\right)$$

$$\Rightarrow f(x) - f(y) = f\left(\frac{x}{y}\right)$$

10. Ans (A,B,C)

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f\left(\frac{x+h}{x}\right)}{\frac{h}{x}}$$

$$= \lim_{h \rightarrow 0} \frac{f\left(1 + \frac{h}{x}\right)}{\frac{h}{x} \cdot x} = \frac{1}{x} f'(1) = \frac{2}{x}$$

$$f'(3) = \frac{2}{3}$$

11. Ans (B,C,D)

$$f(0) = f(1)$$

$$2 = a + 1 - 2a - 1 + 2$$

$$a = 0 \Rightarrow f(x) = x^2 - x + 2$$

12. Ans (A)

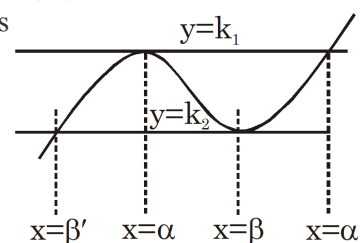
$$f(x) = k_1 \text{ has roots } \alpha, \alpha'$$

$$f(x) = k_2 \text{ has}$$

$$\text{roots } \beta, \beta'$$

$$2\alpha + \alpha' = \frac{-1}{a}$$

$$2\beta + \beta' = \frac{-1}{a}$$



$$2(\alpha + \beta) + \alpha' + \beta' = \frac{-2}{a} \quad \dots(i)$$

$$f'(x) = 0 \Rightarrow 3x^2a + 2x - (2a + 1) = 0$$

$$\therefore \alpha + \beta = \frac{-2}{3a} \quad \therefore \alpha' + \beta' = \frac{-2}{a} + \frac{4}{3a} = \frac{-2}{3a}$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (0.50)**

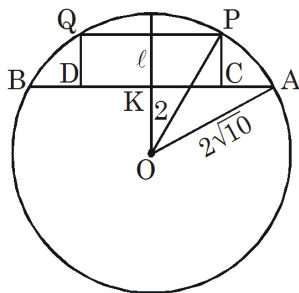
$$\begin{aligned}
 I &= \int \frac{x^{16} \cdot x}{(1+x^2)^{10}} dx & 1+x^2 &= t \\
 &= \frac{1}{2} \int \frac{(t-1)^8}{t^{10}} dt & x dx &= \frac{1}{2} dt \\
 &= \frac{1}{2} \int \left(1 - \frac{1}{t}\right)^8 \cdot \frac{1}{t^2} dt & \text{Put } 1 - \frac{1}{t} &= y \\
 &= \frac{1}{2} \int y^8 dy & \frac{1}{t^2} dt &= dy \\
 &= \frac{1}{18} \cdot \left(\frac{x^2}{1+x^2}\right)^9 + C \\
 \therefore m &= 9, n = 18 \\
 \therefore \frac{m}{n} &= \frac{1}{2}
 \end{aligned}$$

2. **Ans (6.00)**

$$AB = 12 \Rightarrow KA = 6$$

$$\text{Let } PC = \ell$$

$$OK = 2$$



$$PQ = 2\sqrt{40 - (2 + \ell)^2}$$

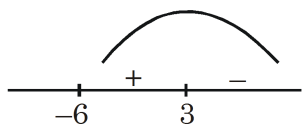
Area of rectangle

$$A = \ell \cdot 2\sqrt{36 - \ell^2 - 4\ell}$$

$$\frac{dA}{d\ell} = 2 \left\{ \frac{\ell(-\ell-2)}{\sqrt{36 - \ell^2 - 4\ell}} + \sqrt{36 - \ell^2 - 4\ell} \right\}$$

$$\Rightarrow (-\ell^2 - 2\ell + 36 - \ell^2 - 4\ell) \cdot \frac{2}{\sqrt{36 - \ell^2 - 4\ell}}$$

$$\Rightarrow -(\ell + 6)(\ell - 3) \cdot \frac{2}{\sqrt{36 - \ell^2 - 4\ell}}$$



$$A_{(\max)} \text{ at } \ell = 3 \Rightarrow A_{\max} = 6\sqrt{15}$$

$$\frac{\lambda}{\sqrt{15}} = 6$$

$$A_{\max} = 6\sqrt{15}$$

3. **Ans (8.00)**

$$f(x) = \frac{195x(x^2 + 4)}{9x^4 + 97x^2 + 144} = \frac{13(5x)(3x^2 + 12)}{(3x^2 + 12)^2 + (5x)^2}$$

$$= \frac{13}{\frac{3x^2+12}{5x} + \frac{5x}{3x^2+12}} \text{ for } x \neq 0$$

$$\text{Then } \left| \frac{3x^2 + 12}{5x} \right| \geq \frac{12}{5}$$

$$\text{Hence } |f(x)| \leq \frac{60}{13}$$

$$a - 4b = 8$$

4. **Ans (4.58 to 4.59)**

$$\int \left(\frac{\sin x \cos x}{1 + \sin x + \cos x} \right)^2 dx = \frac{1}{4} \int \left(\frac{2 \sin x \cos x}{1 + \sin x + \cos x} \right)^2 dx$$

$$\frac{1}{4} \int \left(\frac{1 + 2 \sin x \cos x - 1}{1 + \sin x + \cos x} \right)^2 dx = \frac{1}{4} \int \left(\frac{(\sin x + \cos x)^2 - 1}{1 + \sin x + \cos x} \right)^2 dx$$

$$\frac{1}{4} \int \frac{(\sin x + \cos x + 1)^2 (\sin x + \cos x - 1)^2}{(1 + \sin x + \cos x)^2} dx$$

$$\frac{1}{4} \int (\sin x + \cos x - 1)^2 dx$$

$$= \frac{1}{4} \int (2 + 2 \sin x \cos x - 2 \sin x - 2 \cos x) dx$$

$$\Rightarrow \frac{1}{4} \int (2 + \sin 2x - 2 \sin x - 2 \cos x) dx$$

$$\Rightarrow \frac{x}{2} - \frac{\cos 2x}{8} + \frac{1}{2}(\cos x - \sin x) + \delta$$

$$\Rightarrow \frac{x}{2} - \frac{\cos 2x}{8} + \frac{\cos(x + \frac{\pi}{4})}{\sqrt{2}} + \delta$$

$$\alpha = 2, \beta = -8, \gamma = \sqrt{2}$$

$$|\alpha + \beta + \gamma| = 6 - \sqrt{2} = 6 - 1.414$$

$$= 4.586$$

5. **Ans (0.03)**

$$f(x) = \max\{x, x^2\} = \begin{cases} x & ; x \in [0, 1] \\ x^2 & ; x \in (1, \infty) \end{cases}$$

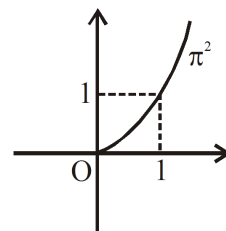
$$\Rightarrow f \circ f(x) = \begin{cases} f(x) & ; 0 \leq f(x) \leq 1 \\ f^2(x) & ; f(x) > 1 \end{cases}$$

$$= \begin{cases} x & 0 \leq x \leq 1 \\ x^4 & x > 1 \end{cases}$$

$$(f \circ f)^{-1}(x) = x^{1/4} \text{ for } x > 1$$

$$\frac{d}{dx} (f \circ f)^{-1}(x) = \frac{1}{4} x^{-3/4}$$

$$\text{put } x = 16$$



6. **Ans (1.00)**

It is 0° form

$$\text{Let } y = (\sin \theta)^{(\sin \theta - \sin^2 \theta)}$$

$$\ln y = (1 - \sin \theta)(\sin \theta \ln \sin \theta)$$

$$\ln y = \lim_{\theta \rightarrow 0} \frac{\ln \sin \theta}{\cos \theta} \quad (\text{Use L'Hopital and solve})$$

7. **Ans (3.00)**

$$\frac{d^3 y}{dx^3} = \frac{d}{dx} \left(\frac{d}{dx} \left(\frac{dy}{dx} \right) \right)$$

$$= \frac{d}{dx} \left(\frac{d}{dy} \left(\frac{1}{\frac{dx}{dy}} \right) \cdot \frac{dy}{dx} \right)$$

$$= \frac{d}{dx} \left(- \frac{1}{\left(\frac{dx}{dy} \right)^2} \cdot \frac{d^2 x}{dy^2} \cdot \frac{dy}{dx} \right)$$

$$= \frac{d}{dx} \left(- \left(\frac{dy}{dx} \right)^3 \cdot \frac{d^2 x}{dy^2} \right)$$

$$= - \left(\frac{dy}{dx} \right)^2 \left(3 \cdot \frac{d^2 x}{dy^2} \cdot \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx} \right)^2 \cdot \frac{d^3 x}{dy^3} \right)$$

8. **Ans (0.50)**

$$\text{Let } C_1 : y = -2\sqrt{x+2} \quad \dots(1)$$

$$C_2 : y = -x - 4 \quad \dots(2)$$

$\therefore$ Diff. b.t.s w.r.t

x in (1),

$$\left. \frac{dy}{dx} \right|_{(t, -2\sqrt{t+2})} = - \frac{1}{\sqrt{t+2}} = -1$$

$$\Rightarrow t = -1$$

$$\therefore d = \left(\frac{|-1 - 2 + 4|}{\sqrt{2}} \right)^2 = \left(\frac{1}{\sqrt{2}} \right)^2 = \frac{1}{2}$$

